

Gioia Zheng

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Profile

B.Sc. student in Applied Computer Science and Artificial Intelligence at Sapienza University of Rome. Building reproducible retrieval and evaluation systems for large-scale QA pipelines. Interested in research-engineering problems at the intersection of information retrieval, evaluation methodology, and AI infrastructure.

Education

Sapienza University of Rome

Sep 2022 - Present

B.Sc. in Applied Computer Science and Artificial Intelligence

Projects

MS MARCO Generative QA System

github.com/GioiaZheng/msmarco-genqa

Python - Retrieval, Reranking, Generation, Evaluation

- Built an end-to-end retrieve -> rerank -> generate pipeline on MS MARCO (6,980 queries, 8.8M passages), improving Token-F1 from 0.197 to 0.368.
- Reproduced sparse, dense, and cross-encoder retrieval stages on CPU-only infrastructure using BM25, SBERT/FAISS, and MiniLM reranking.
- Built research-engineering infrastructure including CI, config-driven pipelines, experiment manifests, benchmark reporting, optional FastAPI serving, and local / MLflow / W&B tracking.
- Analyzed disagreement between lexical and semantic grounding metrics, revealing evaluation inconsistencies in generated QA outputs.

Handwritten OCR Pipeline

github.com/GioiaZheng/handwritten-ocr-system

PyTorch - CRNN, BiLSTM, CTC

- Designed and trained a CRNN-based handwritten text recognition system combining residual CNN feature extraction, BiLSTM sequence modeling, and CTC decoding.
- Implemented end-to-end training, evaluation, and monitoring pipelines with CER/WER tracking and TensorBoard instrumentation.

Chat Service

github.com/GioiaZheng/go-chat-system

Go Backend + Vue Frontend + SQLite

- Developed a self-contained chat platform with Go backend, Vue frontend, and SQLite persistence packaged as a single deployable artifact.
- Established automated quality gates including OpenAPI validation, testing, and CI-based build verification.

Experience

Web Development Intern

Mar 2026 - Present

Idearia - Rome, Italy

- Developed and maintained PHP / WordPress backend systems for production websites.
- Built automation workflows using n8n to connect forms, content pipelines, and external services.

NLP Algorithm Engineer Intern

Feb 2026 - May 2026

Microsoft - Natural Language Computing (Remote)

- Contributed to retrieval-augmented QA experimentation and evaluation on the MS MARCO benchmark.
- Built and evaluated retrieval, reranking, generation, and analysis components over a 6,980-query benchmark using BM25, FAISS, sentence-transformers, and T5.
- Conducted reproducibility, grounding, and failure-analysis studies for RAG pipelines.

Selected Writing

Retrieval Evaluation Beyond Surface Metrics

Technical note - RAG evaluation, grounding, failure analysis

gioiazheng.github.io/writing/rag-failure-analysis

MS MARCO GenQA Technical Report

Project report - reproducible retrieval, reranking, and generation

gioiazheng.github.io/projects/msmarco-genqa

Skills

Languages	Python, Go, SQL, JavaScript, PHP
AI & Research	Information Retrieval, RAG Evaluation, PyTorch, FAISS, BM25, Sentence Transformers, Cross-Encoder Reranking
Web & Automation	WordPress, n8n, REST APIs, Vue, OpenAPI
Systems	Linux, Docker, GitHub Actions, SQLite
Spoken Languages	Chinese (Native), Italian (Bilingual), English (Professional), German (Beginner)